

## SAFETY DATA SHEET

Creation Date 21-Sep-2009

Revision Date 18-Sep-2024

Revision Number 6

### 1. Identification

**Product Name** Thionyl chloride

**Cat No. :** S37605

**CAS No** 7719-09-7  
**Synonyms** Thionyl dichloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** All other uses.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |              |
|--------------------------------------------------|--------------|
| Acute oral toxicity                              | Category 4   |
| Acute Inhalation Toxicity - Vapors               | Category 3   |
| Skin Corrosion/Irritation                        | Category 1 A |
| Serious Eye Damage/Eye Irritation                | Category 1   |
| Specific target organ toxicity (single exposure) | Category 3   |
| Target Organs - Respiratory system.              |              |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Harmful if swallowed  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
Toxic if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Do not breathe dust/fume/gas/mist/vapors/spray  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Reacts violently with water  
Contact with water liberates toxic gas

### 3. Composition/Information on Ingredients

| Component        | CAS No    | Weight % |
|------------------|-----------|----------|
| Thionyl chloride | 7719-09-7 | <=100    |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

|                                     |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation                          | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| Ingestion                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Most important symptoms and effects | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                |
| Notes to Physician                  | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

|                                  |                                                  |
|----------------------------------|--------------------------------------------------|
| Suitable Extinguishing Media     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. |
| Unsuitable Extinguishing Media   | Do not use water or foam                         |
| Flash Point                      | No information available                         |
| Method -                         | No information available                         |
| Autoignition Temperature         | No information available                         |
| Explosion Limits                 |                                                  |
| Upper                            | No data available                                |
| Lower                            | No data available                                |
| Sensitivity to Mechanical Impact | No information available                         |
| Sensitivity to Static Discharge  | No information available                         |

### Specific Hazards Arising from the Chemical

Reacts violently with water. Contact with water liberates toxic gas. The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Chlorine. Sulfur oxides. Hydrogen chloride gas. Phosgene.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

Health  
3

Flammability  
1

Instability  
0

Physical hazards  
W

## 6. Accidental release measures

|                                      |                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal Precautions                 | Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Wear self-contained breathing apparatus and protective suit. |
| Environmental Precautions            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                          |
| Methods for Containment and Clean Up | Do not expose spill to water. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.                                                      |

## 7. Handling and storage

|          |                                                                                                                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not allow contact with water. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.**

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from water or moist air. Incompatible Materials. Acids. Bases. Water. Strong oxidizing agents. Alcohols. Amines.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component        | ACGIH TLV        | OSHA PEL                                                           | NIOSH                                          | Mexico OEL (TWA) |
|------------------|------------------|--------------------------------------------------------------------|------------------------------------------------|------------------|
| Thionyl chloride | Ceiling: 0.2 ppm | (Vacated) Ceiling: 1 ppm<br>(Vacated) Ceiling: 5 mg/m <sup>3</sup> | Ceiling: 1 ppm<br>Ceiling: 5 mg/m <sup>3</sup> | Ceiling: 0.2 ppm |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                             |
|----------------------------------------|-----------------------------|
| Physical State                         | Liquid                      |
| Appearance                             | Light yellow                |
| Odor                                   | Strong, pungent             |
| Odor Threshold                         | 0.9 ppm                     |
| pH                                     | Strongly acidic             |
| Melting Point/Range                    | -105 °C / -157 °F           |
| Boiling Point/Range                    | 76 °C / 168.8 °F @ 760 mmHg |
| Flash Point                            | No information available    |
| Evaporation Rate                       | No information available    |
| Flammability (solid,gas)               | Not applicable              |
| Flammability or explosive limits       |                             |
| Upper                                  | No data available           |
| Lower                                  | No data available           |
| Vapor Pressure                         | 124 mbar @ 20 °C            |
| Vapor Density                          | 4.1                         |
| Specific Gravity                       | 1.640                       |
| Solubility                             | No information available    |
| Partition coefficient; n-octanol/water | No data available           |
| Autoignition Temperature               | No information available    |
| Decomposition Temperature              | 140 °C                      |
| Viscosity                              | 0.6 mPa.s at 20 °C          |

Molecular Formula Cl<sub>2</sub> O S  
Molecular Weight 118.97

## 10. Stability and reactivity

**Reactive Hazard** Yes

**Stability** May react with metals and lead to the formation of flammable hydrogen gas. Contact with water liberates toxic gas. Moisture sensitive.

**Conditions to Avoid** Exposure to moist air or water. Exposure to light. Incompatible products.

**Incompatible Materials** Acids, Bases, Water, Strong oxidizing agents, Alcohols, Amines

**Hazardous Decomposition Products** Chlorine, Sulfur oxides, Hydrogen chloride gas, Phosgene

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing. Reacts violently with water.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component        | LD50 Oral                | LD50 Dermal | LC50 Inhalation               |
|------------------|--------------------------|-------------|-------------------------------|
| Thionyl chloride | LD50 = 324 mg/kg ( Rat ) | Not listed  | LC50 = 2.717 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component        | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|-----------|------------|------------|------------|------------|------------|
| Thionyl chloride | 7719-09-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available. Do not empty into drains.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1836  
Hazard Class 8  
Packing Group I

### TDG

UN-No UN1836  
Hazard Class 8  
Packing Group I

### IATA

UN-No UN1836  
Proper Shipping Name THIONYL CHLORIDE, FORBIDDEN FOR IATA TRANSPORT  
Hazard Class 8  
Packing Group I

### IMDG/IMO

UN-No UN1836  
Proper Shipping Name THIONYL CHLORIDE  
Hazard Class 8  
Packing Group I

## 15. Regulatory information

### United States of America Inventory

| Component        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Thionyl chloride | 7719-09-7 | X    | ACTIVE                                        | -                           |

### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component        | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Thionyl chloride | 7719-09-7 | X   | -    | 231-748-8 | X     | X    | X    | X    | X     | KE-33794 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** **OSHA - United States Occupational Safety and Health Administration**

| Component        | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|------------------|----------------------------------|----------------------------|
| Thionyl chloride | -                                | TQ: 250 lb                 |

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

### **U.S. State Right-to-Know Regulations**

| Component        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------------|---------------|------------|--------------|----------|--------------|
| Thionyl chloride | X             | X          | X            | -        | X            |

#### **U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

#### **U.S. Department of Homeland Security**

This product contains the following DHS chemicals:  
**Legend** - STQs = Screening Threshold Quantities, APA = A placarded amount

| Component        | DHS Chemical Facility Anti-Terrorism Standard |
|------------------|-----------------------------------------------|
| Thionyl chloride | APA                                           |

### Other International Regulations

**Mexico - Grade** No information available

### **Authorisation/Restrictions according to EU REACH**

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous | REACH Regulation (EC 1907/2006) article 59 - Candidate List of |
|-----------|--------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------|
|-----------|--------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------|

|                  |           |   | Substances                                                          | Substances of Very High Concern (SVHC) |
|------------------|-----------|---|---------------------------------------------------------------------|----------------------------------------|
| Thionyl chloride | 7719-09-7 | - | Use restricted. See entry 75.<br>(see link for restriction details) | -                                      |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component        | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Thionyl chloride | 7719-09-7 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Thionyl chloride | 7719-09-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Health, Safety and Environmental Department  
Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**